

Analog IC Design Implementation and Verification

Week 6 – Noise Analysis

Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1

Revision 1.0

Estimated time: 1 Week

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Module Objectives

In this module, you will learn how to

- Analyze circuits to characterize sources of noise
- Predict noise behavior
- Make design trade-offs to reduce noise
- Calculate and apply the transfer function of a circuit to model signal behavior
- Reference noise to different points throughout the circuit
- Describe the sources of noise and the statistical measures of noise in amplifiers
- Compare the characteristics of shot noise, thermal noise, and flicker noise
- Compare noise models for resistors, diodes, bipolar junction transistors, and MOS transistors



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Introduction: Noise Statistics

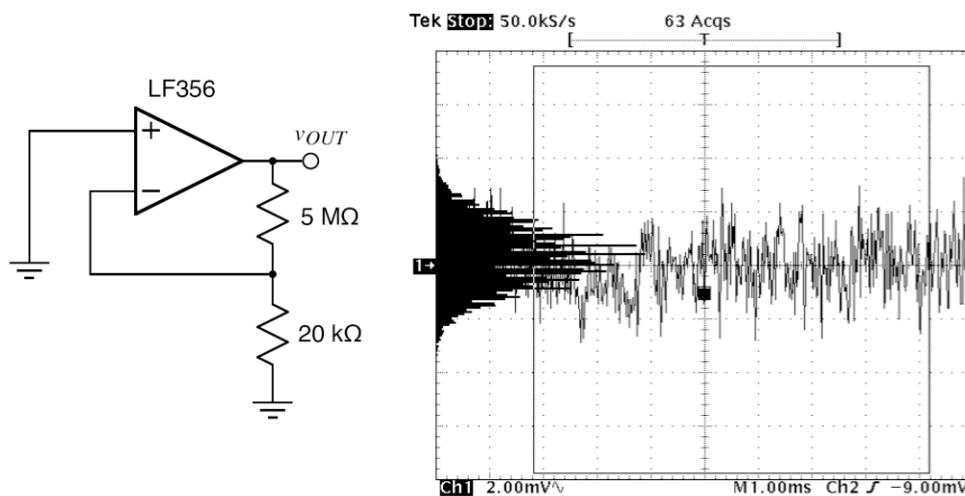
- Noise in integrated circuits
- Noise characterization
 - Statistical measures
 - Time domain measures
- Interpretation of noise statistics



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Introduction: Noise

Ground input of high-gain opamp circuit: *What do you see on v_{OUT} ?*



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What Is Noise?

One definition of *noise* is everything *except* the desired signal.

- Interference
 - Coupling from external sources (example: crosstalk)
 - Not random (can, in principle, be eliminated – better shielding, layout)
- For purposes of this module, noise consists of:
 - *Random fluctuations within the device inherent to its operation*
- Why is noise a problem?
 - Imposes a lower limit to the amplitude of the useful signal that can be processed by a circuit without seriously affecting its quality.
 - Imposes an upper limit on the useful gain of an amplifier, for example, a very high-gain amplifier can saturate by amplifying the noise generated in the input stages.
 - Important in low-noise amplifier design, radio receiver design, transducer interface circuitry, and so forth.
 - Critical not only in receivers but also often in transmitting circuits as well because they act as disturbers to all potential receivers.



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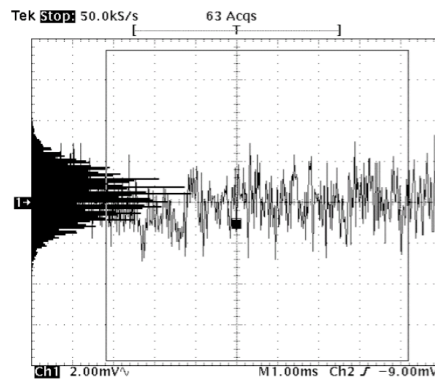
Description: Noise

Sine wave: described completely by

- Amplitude
- Phase
- Frequency

How to describe noise waveforms?

- Must use statistical measures



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Noise Statistics

Noise generation is a random (stochastic) process. It must use statistical properties to characterize noise waveforms:

- Mean
- Variance
- Autocorrelation function
- Power spectral density

Assume noise processes are:

- Stationary, that is, their statistical properties are time invariant
- Ergodic, that is, their ensemble statistical properties are equal to the statistical properties averaged over time

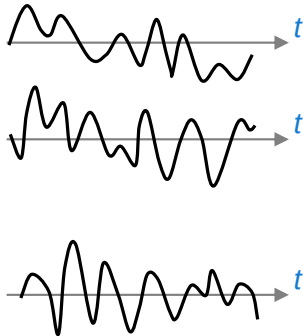
What is an ensemble?



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Random Processes

Formal definition of a random process begins with its (infinite) ensemble of possible waveform realizations:

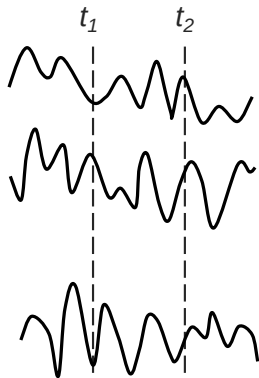


Multiple realizations of the same random signal

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Stationarity

Formal definition of a random process begins with its (infinite) ensemble of possible waveform realizations:



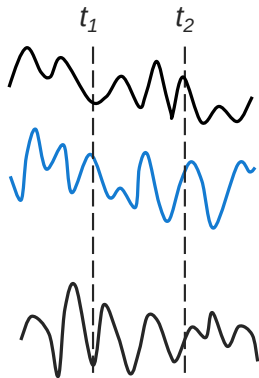
Statistics (for example, average value) are defined at each time $t_1, t_2, \dots$ instant over the ensemble of all waveforms in the random process.

If these statistics do not change in time, the random process is called *stationary*.

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Ergodicity Assumption

In the real world, the infinite number of realizations of a stochastic process are not accessible.

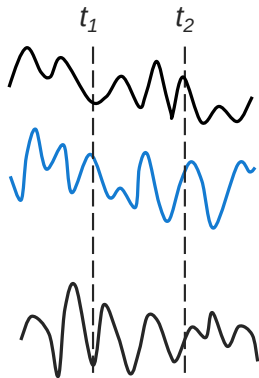


You must evaluate the properties of a random process based on a *single realization* of it.

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Ergodicity

Formal definition of a random process is defined by its (infinite) ensemble of possible realizations.



If statistics averaged over time for any waveform equal the ensemble average statistics, then the random process is *ergodic*.

Cannot be proven. Must be assumed.

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Expectation Operator

In the following expressions, $f(x)$ represents the probability density function of the random variable X :

- Expected value of X is defined as:

$$E\{X\} = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

- Example: quantization noise of ADC, uniform from $-1/2$ to $+1/2$



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Mean of a Random Variable

Expected value is also called the mean:

$$\mu_x = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

- Different random variables can have the same mean but have very differently.
- How to describe differences in variation?



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Variance of a Random Variable

The expected value of distance squared from the mean: The variance of a stationary random signal $x(t)$ is defined as:

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 f(x) dx$$



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Standard Deviation of a Random Variable

The expected value of distance squared from the mean: The variance of a stationary random signal $x(t)$ is defined as:

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 f(x) dx$$

- Variance has units of quantity squared.
- If you want measurements with the same units, use standard deviation.

$$\sigma_x = \sqrt{\sigma_x^2}$$



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Statistical Properties of Signals

- Use ergodicity assumption to approximate statistics from time averages.
- Ideally, this is infinite time; however, use “long enough” record length.



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DC Component and Power of a Signal

The average value (DC component) of a signal $x(t)$ is:

$$\bar{x} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt$$

The average power delivered to a 1Ω resistor by a signal $x(t)$ is:

$$\overline{x^2} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)^2 dt$$

The average power of the varying component of the signal $x(t)$ is:

$$\overline{(x - \bar{x})^2} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T (x(t) - \bar{x})^2 dt$$



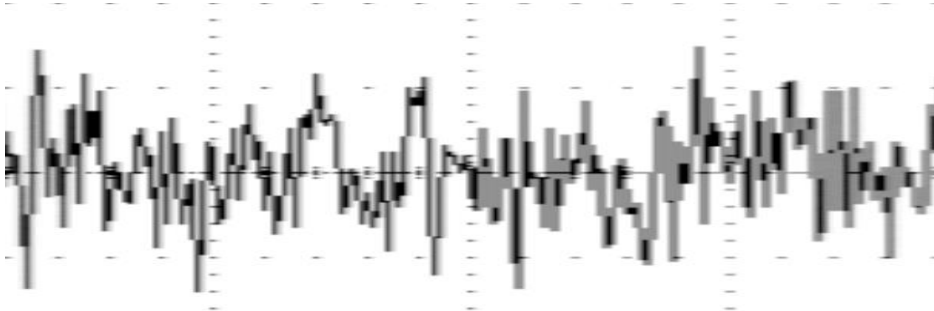
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How Does Signal Change With Time?

Does knowledge of signal value at t_1 tell us anything about the signal at t_2 ?

If t_1 close to t_2 , maybe.

If t_1 far from t_2 , probably not.



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Autocorrelation Function

The autocorrelation function of a stationary random process (signal) $x(t)$ is defined as the expected value (average) of the signal multiplied by a version of itself delayed by τ :

$$R_x(\tau) = E\{x(t)x(t + \tau)\}$$



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The Time-Domain Autocorrelation Function

Because all waveforms or all time are not accessible, use ergodicity:

- The time-domain autocorrelation function of a signal $x(t)$ is:

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t-\tau)dt$$



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Average Power in a Signal

Substitute $\tau=0$ in autocorrelation definition:

$$R_x(0) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)x(t)dt$$

This is just the average power of the signal $x(t)$; can also be written as

$$P_{av}(x) = \overline{x^2} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t)^2 dt = R_x(0)$$



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Interpretation of Noise Statistics

- For an ergodic signal, its time average is equal to its statistical mean:

$$\bar{x} = \mu_x,$$

- The average power of its varying component is equal to its variance:

$$\overline{(x - \bar{x})^2} = \sigma_x^2,$$

and the time-domain autocorrelation function is equal to its statistical counterpart.

- The average power of the signal $x(t)$ also can be computed by using its power spectral density (PSD or power spectrum):

$$P_{av}(x) = R_x(0) = \int_{-\infty}^{\infty} S_x(f) df$$



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Autocorrelation Function and Power Spectral Density

The power spectral density (PSD) $S_x(f)$ of a random process and the autocorrelation function $R_X(t)$ of a random process are related by a Fourier transform relationship (Wiener-Khinchine theorem):

$$R_x(t) = \int_{-\infty}^{\infty} S_x(f) e^{j \cdot 2\pi f t} df \quad \Leftrightarrow \quad S_x(f) = \int_{-\infty}^{\infty} R_x(t) e^{-j \cdot 2\pi f t} dt$$



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Autocorrelation Function and Power Spectral Density (continued)

How signal is correlated over time

$\Leftrightarrow$ How power is distributed over frequency

$$R_x(t) = \int_{-\infty}^{\infty} S_x(f) e^{j \cdot 2\pi f t} df$$

$$\Leftrightarrow S_x(f) = \int_{-\infty}^{\infty} R_x(t) e^{-j \cdot 2\pi f t} dt$$



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Units

How signal is correlated over time

⇔ How power is distributed over frequency

$$R_x(t) = \int_{-\infty}^{\infty} S_x(f) e^{j \cdot 2\pi f t} df$$

$$S_x(f) = \int_{-\infty}^{\infty} R_x(t) e^{-j \cdot 2\pi f t} dt$$

Integral of PSD over frequency = Average noise power



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Interpretation of Noise Statistics (continued)

- For analog designers these measures are usually most important.
- Power spectral density (PSD), for example, in $\text{nV}/\sqrt{\text{Hz}}$ or $\text{pA}/\sqrt{\text{Hz}}$.
- Integral of PSD over frequency gives the RMS value in a certain bandwidth. This can be directly compared to the signal. The ratio between signal and noise, or SNR, gives you a good indication of the noise level in your circuit.
- PSD is the usual output of an analog circuit simulator. Often a table output showing the contribution of each circuit component is also available (noise summary).
- The previous concepts are appropriate for time-invariant systems.
For time-variant circuits, such as oscillators, mixers, and switched-cap designs, simulation with SpectreRF® can provide equivalent noise metrics.



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Summary: Noise Statistics

- Noise is present in virtually all circuits. It is a random process and is therefore defined by its infinite number of realizations.
- Because noise is statistical in nature, standard measurements, such as mean, variance, and so forth, can also be applied to noise.
- For analog designers, certain parameters are especially important such as PSD and SNR.



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Introduction: Noise Types

- Shot noise
- Thermal noise
- Flicker noise



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Shot Noise

- Shot noise is associated with the flow of carriers across a barrier $\Rightarrow$
 - Independent, random events
- Example: direct current flow across a pn junction



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Shot Noise (continued)

- Power spectral density of the shot noise current is:

$$S_i(f) = 2qI_D = i_n^2, \quad (\text{A}^2/\text{Hz})$$

where q is the elementary charge and I_D is the direct current flowing through the pn junction.

- Noise density is:

$$i_n = \sqrt{2qI_D} \quad (\text{A}/\sqrt{\text{Hz}})$$

- Because the power spectral density is constant in frequency, the shot noise is white noise.
- The power delivered by the shot noise current over a frequency bandwidth of Δf into a 1Ω resistor is:

$$\sigma_i^2 = \overline{i^2} = 2qI_D \Delta f$$



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Shot Noise: Numerical Example

What is the rms value of the shot noise current associated to a forward biased pn junction with a direct current of 1 mA, computed within a bandwidth of 1 MHz?

- Noise density:

$$q = 1.6 \cdot 10^{-19} \quad I_D = 10^{-3}$$

$$i_n^2 = 2qI_D = 3.2 \cdot 10^{-22} \text{ A}^2 / \text{Hz} \quad i_n = \sqrt{2qI_D} = 17.9 \text{ pA} / \sqrt{\text{Hz}}$$

- Integrate density over 1MHz bandwidth:

$$\Delta f = 10^6$$

$$\overline{i_n^2} = 2qI_D \Delta f = 3.2 \cdot 10^{-16} \text{ A}^2 = \sigma_i^2$$

$$i_{n,rms} = \sqrt{\overline{i_n^2}} = 18 \text{ nA}_{rms}$$



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Thermal Noise

Thermal noise is associated with random thermal motion of electrons within the physical body of a resistor. The power spectral density of the thermal noise is proportional with the absolute temperature.

For a resistor, thermal noise can be modeled as a voltage PSD:

$$S_v(f) = 4kTR = e_{nR}^2 \text{ V}^2/\text{Hz}$$

$$e_{nR} = \sqrt{4kTR} \text{ V}/\sqrt{\text{Hz}}$$

- k = Boltzmann's constant 1.38E-23 [J/K]
- T = absolute temperature in [K]
- R = resistor value

Because the power spectral density is constant in frequency, the thermal noise is also white noise.



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Thermal Noise (continued)

Or, with Thevenin-Norton equivalence, as a current PSD:

$$S_i(f) = \frac{4kT}{R} = i_{nR}^2 \text{ A}^2/\text{Hz}$$

$$i_{nR} = \sqrt{\frac{4kT}{R}} \text{ A}/\sqrt{\text{Hz}}$$

The two models are equivalent: use whichever gives the easier analysis.



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Thermal Noise: Numerical Example

The power associated to the thermal-noise voltage or current over a frequency bandwidth of Δf is?

What is the rms value of the thermal noise voltage associated with a $1\text{k}\Omega$ resistor within a bandwidth of 1MHz at 27°C ?

$$k = 1.38E - 23 [J / K] \quad T = 300 [K] \quad R = 1000 [\Omega]$$

$$e_{nR}^2 = 4kTR = 1.66E - 17 [V^2 / Hz] \quad e_{nR} = \sqrt{4kTR} = 4.1nV / \sqrt{Hz}$$

Integrate density over 1MHz bandwidth:

$$\Delta f = 10^6$$

$$\sigma_v^2 = 4kTR\Delta f = 1.66E - 11 V^2$$

$$v_{n,rms} = \sqrt{\sigma_v^2} = 4.1\mu V_{rms}$$



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Flicker Noise

- Flicker noise is generally believed to be associated to surface traps in semiconductive materials. These traps capture and release the carriers in a random manner.
- Flicker noise is present in diodes, bipolar junction transistors, and MOS transistors.
- The flicker noise current power spectral density is:

$$S_i(f) = K_1 \frac{I^a}{f^b} = \frac{K_2}{f^b}$$

where K_1 is an empirical constant that is device specific, $0.5 < a < 2$, $b \approx 1$, and I is the direct current of the device.

- Because the power spectral density is inversely proportional to the frequency, the flicker noise is also called *1/f noise*.

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Flicker Noise (continued)

- $1/f$ noise is important for low-frequency applications.
- K_1 can vary orders of magnitude from one device to another. K_1 can also vary widely for different transistors on the same wafer due to the dependence on defects.
- Assuming parameter $b=1$, the power delivered into a 1Ω resistor by the flicker noise current over a frequency bandwidth (f_1, f_2) is:

$$\sigma_i^2 = K_1 I^a \ln\left(\frac{f_2}{f_1}\right)$$

- If $f_2 - f_1 = \Delta f \ll f_1$, one can approximate:

$$\sigma_i^2 \approx \frac{K_1 I^a}{f_1} \Delta f$$



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Summary: Noise Types

Three types caused by the direct current through a pn junction are:

- **Thermal noise:** Caused by the temperature-induced random motions of electrons in a resistor.
- **Flicker noise:** Believed to be associated with surface traps in semiconductive materials, of which noise is of particular importance in analog IC design.
- **Shot noise:** Capture and release the carriers in a random manner.



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Introduction: Representation of Noise in Circuits

- Noise models
 - Resistor
 - Diode
 - Bipolar junction transistor
 - MOS transistor
- Noise analysis
- Equivalent-input referred noise generators



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Noise Models

- Noise models are small-signal models.
- A noisy device is represented by its noiseless small-signal model augmented with independent voltage or current sources, one for each noise generation process.
- All individual noise generators are uncorrelated.
- The noise generators are described by the power of the noise signals they represent.
- Because noise is a random quantity, the polarity used for voltage and current noise sources is unimportant. Nevertheless, once a polarity is chosen, it must be retained throughout the analysis of the circuit.



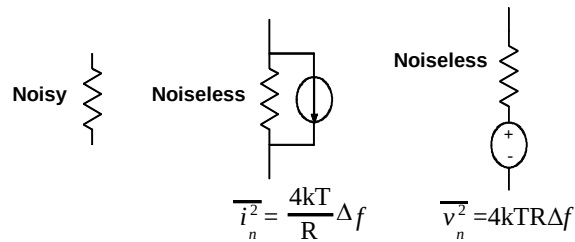
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Resistor Noise Model

The noise model of a resistor is represented by an ideal noiseless resistor and one of these:

- Series-noise voltage source
- Parallel-noise current source

The following diagrams represent thermal noise.

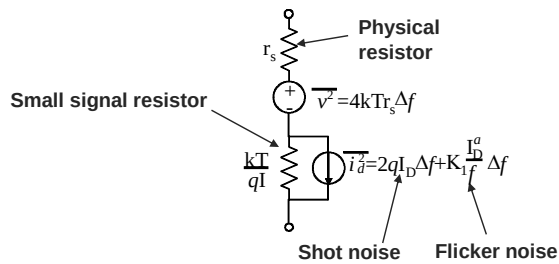


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Diode Noise Model

The noise model of a forward-biased diode contains:

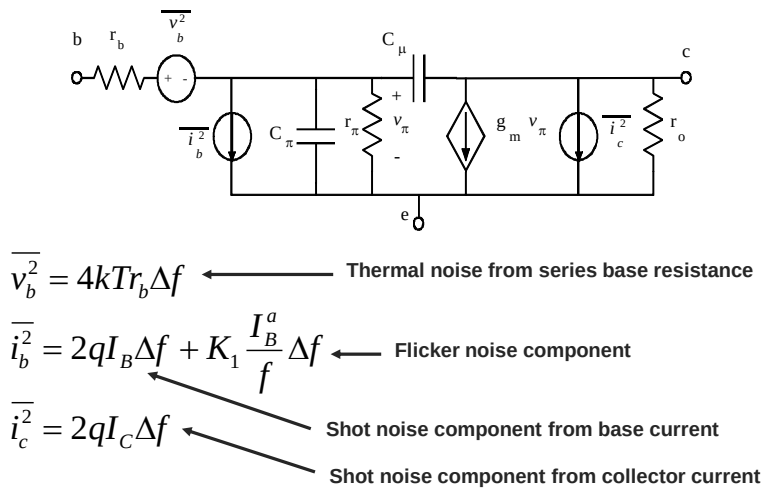
- A physical series resistor, r_s , and the thermal-noise generator associated with it.
- A small-signal resistor and the noise generators associated to the flicker noise and shot noise.
- A small-signal resistor does not generate thermal noise.



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BJT Noise Model

The noise model of a BJT operating in the forward active region is shown in the figure:



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Noise Analysis

- Noise analysis: Add the effect of all noise sources to obtain the total noise power.
- Because noise signals are random, the average power of a sum of two signals $x_1(t)$ and $x_2(t)$ is:

$$P_{av}(x_1 + x_2) = \overline{(x_1 + x_2)^2} = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T (x_1(t) + x_2(t))^2 dt =$$

$$= P_{av}(x_1) + P_{av}(x_2) + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T x_1(t) \cdot x_2(t) dt$$

- The last term is the cross-correlation function between $x_1(t)$ and $x_2(t)$ and indicates how similar are the two signals.
- Because noise from different sources are independent, such noise signals are uncorrelated. Therefore:

$$P_{av}(x_1 + x_2) = \overline{(x_1 + x_2)^2} = \overline{x_1^2} + \overline{x_2^2} = P_{av}(x_1) + P_{av}(x_2)$$



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Output Noise

What happens when a noise signal drives a linear, time-invariant circuit described by a transfer function $H(s)$?

- **Theorem:** The power-density spectrum of the output signal $y(t)$ of a linear, time-invariant system $H(s)$ is:

$$S_y(f) = S_x(f) \cdot |H(f)|^2$$

where $S_x(f)$ is the power density spectrum of the input signal $x(t)$, and $H(f) = H(s)/s = 2j\pi f$.

- The average power of the output noise signal is:

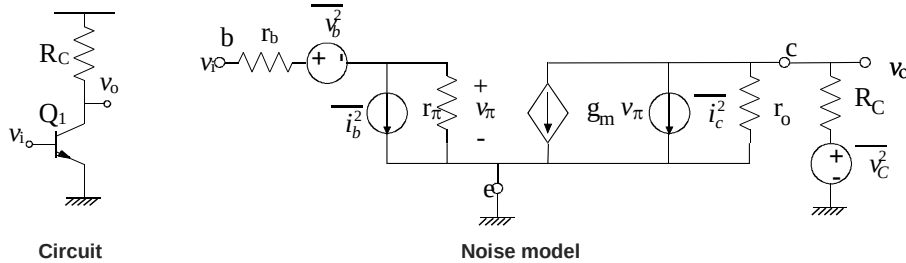
$$P_{av}(y) = \int_{-\infty}^{\infty} S_y(f) df = \int_{-\infty}^{\infty} S_x(f) |H(f)|^2 df$$



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Noise Analysis: Numerical Example

- Compute the total output noise voltage of a common emitter stage in the 10 Hz to 1 MHz bandwidth:



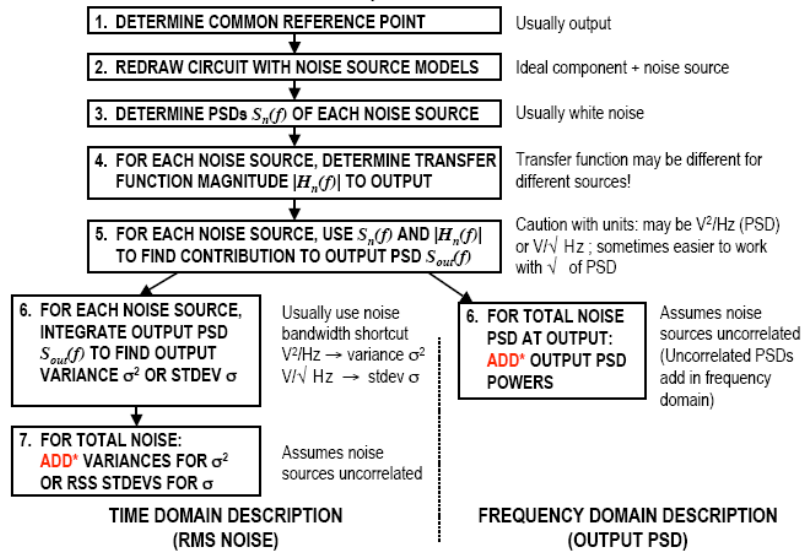
- The transfer functions of the four noise generators are:

$$H_{v_b}(s) = \frac{g_m r_\pi}{r_\pi + r_b} (r_o \parallel R_C), \quad H_{i_b}(s) = g_m (r_\pi \parallel r_b) (r_o \parallel R_C),$$

$$H_{i_c}(s) = r_o \parallel R_C, \quad H_{v_c}(s) = \frac{r_o}{r_o + R_C}$$

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Noise Analysis Procedure



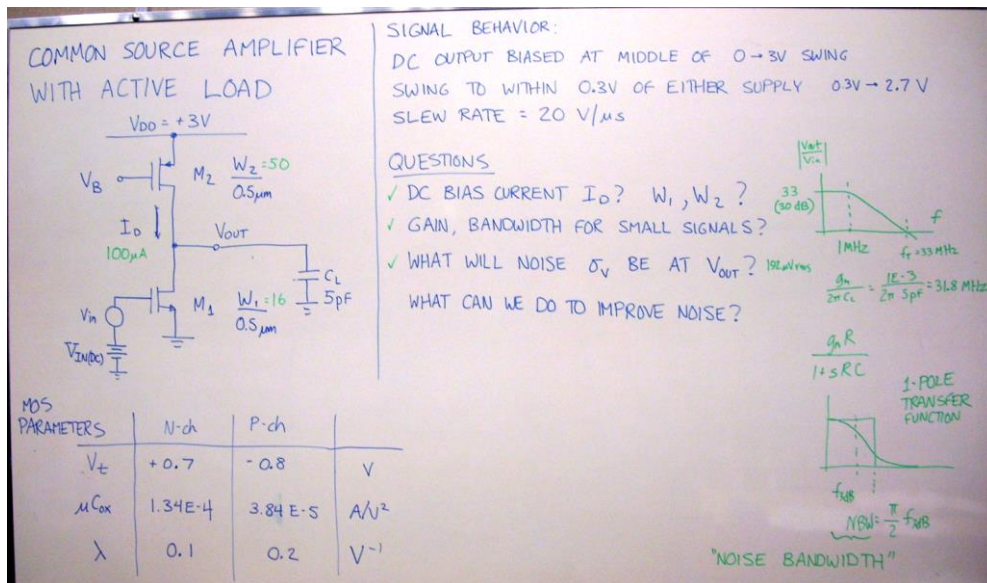
* "ADD" $\rightarrow$ IMPLIES SUPERPOSITION $\rightarrow$ REQUIRES LINEARITY $\rightarrow$ NOISE IS SMALL SIGNAL $\rightarrow$ SUPERPOSITION VALID

Reference: http://ece.wpi.edu/~mcneill/handouts/Noise_procedure.pdf



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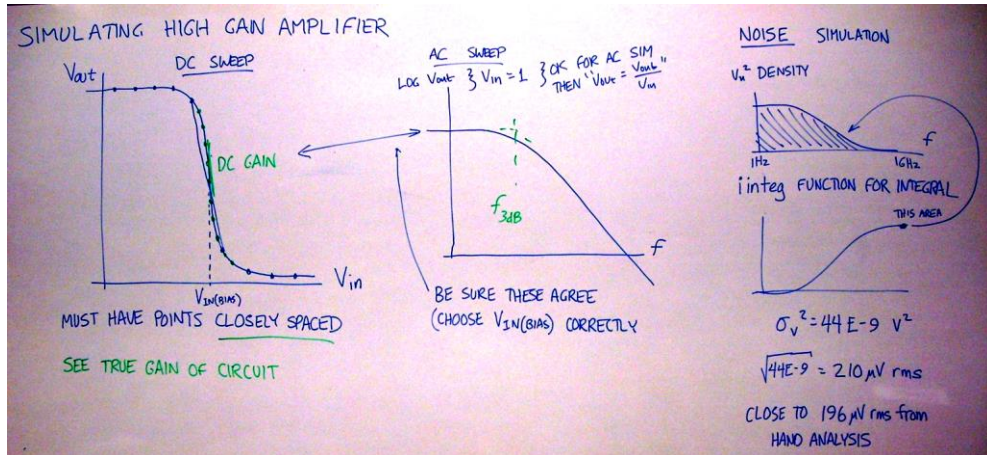
Noise Analysis Procedure: Design Example



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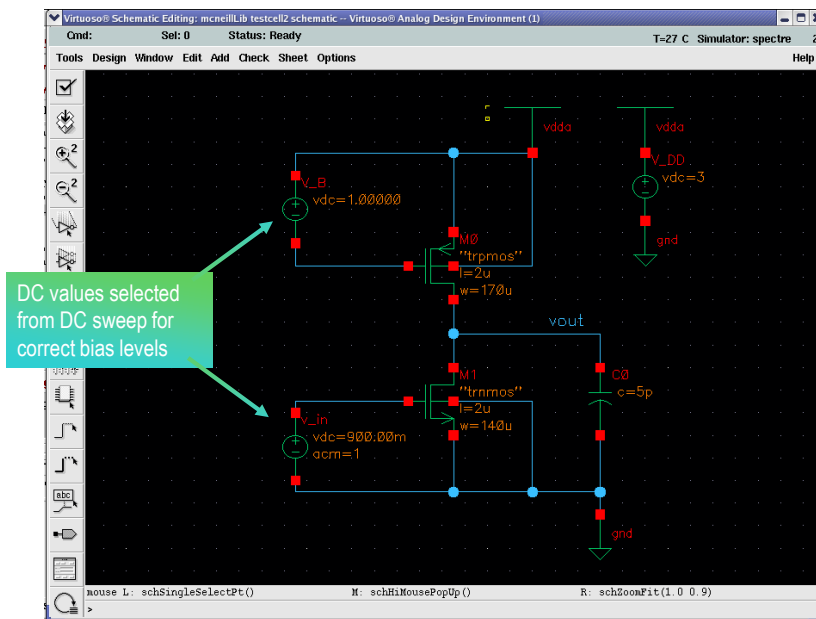
Noise Analysis: Simulation Strategy

- Relate results from DC, AC, noise simulations.
- Simulation total rms noise results are within 5% of hand analysis.



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Noise Analysis: Simulation Schematic

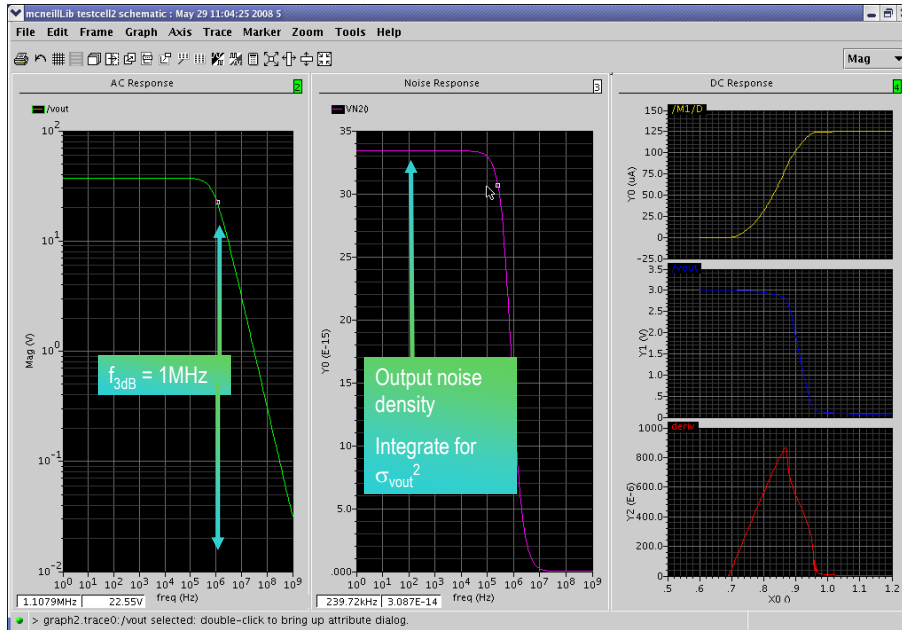


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Noise Analysis: Simulation Results



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Noise Analysis: Numerical Example (continued)

- The power-spectrum density of the output noise voltage is:

$$S_{v_{on}}(f) = 4kTr_b \frac{[g_m r_\pi (r_o \parallel R_C)]^2}{(r_b + r_\pi)^2} + 2qI_C (r_o \parallel R_C)^2 + \\ + \left(2qI_B + K \frac{I_B^a}{f} \right) [g_m (r_\pi \parallel r_b) (r_o \parallel R_C)]^2 + 4kTR_C \frac{r_o^2}{(r_o + R_C)^2}$$

- The total output noise voltage power in the (f_1, f_2) bandwidth results:

$$\overline{v_{on}^2} = \left\{ \left(2qI_B + \frac{4kT}{r_b} \right) [g_m (r_\pi \parallel r_b)]^2 + 2qI_C + \frac{4kT}{R_C} \right\} (r_o \parallel R_C)^2 (f_2 - f_1) + KI_B^a \ln \left(\frac{f_2}{f_1} \right) [g_m (r_\pi \parallel r_b) (r_o \parallel R_C)]^2$$

- Assume $I_C=1$ mA, $V_A=100$ V, $\beta_F=100$, $r_b=100\Omega$, $R_C=2$ k Ω , $K=10^{-14}$, $a=1$.

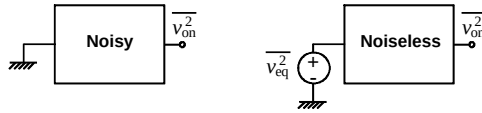
$$g_m \approx 40\text{mS}; \quad r_\pi = 2.5\text{k}\Omega; \quad r_o = 100\text{k}\Omega \\ H_{v_b} \approx 77; \quad H_{i_b} \approx 8\text{k}\Omega; \quad H_{i_c} \approx 2\text{k}\Omega; \quad H_{v_c} \approx 1 \quad \Rightarrow \sqrt{\overline{v_{on}^2}} = 30.22\mu\text{V}$$



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What Are Equivalent-Input Noise Generators?

- The minimum level of the input signal that can be processed without affecting its quality is set by the equivalent-input referred noise.
- **Definition:** The input-referred noise *voltage* represents the value of an equivalent noise voltage applied to the input of a noiseless circuit that produces the same amount of output noise as the circuit including all its noise sources with its input tied to ground.



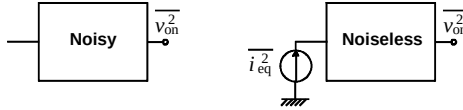
- The PSD of the input referred noise voltage can be computed as:

$$S_{v_{eq}}(f) = \frac{S_{v_{on}}(f)}{|H_v(f)|^2}$$

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Equivalent-Input Noise Generators (continued)

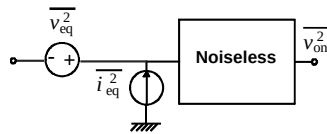
Current: The input-referred noise current represents the value of an equivalent-noise current applied to the input of a noiseless circuit that produces the same amount of output noise as the circuit including all its noise sources with its input left in the open circuit.



- The PSD of the input-referred noise current can be computed as:

$$S_{i_{eq}}(f) = \frac{S_{v_{on}}(f)}{|H_i(f)|^2}$$

- Complete representation of a noisy circuit with equivalent input noise generators:



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Noise in Feedback Amplifier

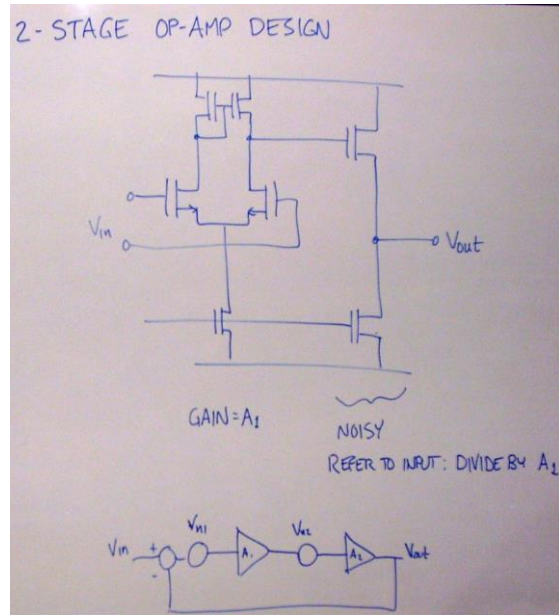
Two-stage amplifier

- Stage 1: noise v_{n1} , gain A_1
- Stage 2: noise v_{n2} , gain A_2
- Open-loop: output noise $v_{n(out,OL)}$

$$v_{out(OL)} = \underbrace{A_1 A_2}_{GAIN} v_{in} + \underbrace{A_1 A_2 v_{n1} + A_2 v_{n2}}_{v_{n(out,OL)}}$$

Input referred: divide by gain

$$v_{n(in,OL)} = v_{n1} + \frac{1}{A_2} v_{n2}$$



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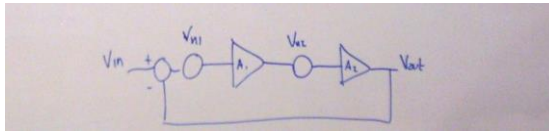
Noise in Feedback Amplifier (continued)

Closed-loop: output noise $v_{n(out,CL)}$

$$v_{out(CL)} = A_2(v_{n2} + A_1(v_{n1} + v_{in} - v_{out(CL)}))$$
$$v_{out(CL)} = \underbrace{\frac{A_1 A_2}{1 + A_1 A_2} v_{in}}_{GAIN} + \underbrace{\frac{A_1 A_2}{1 + A_1 A_2} v_{n1} + \frac{A_2}{1 + A_1 A_2} v_{n2}}_{v_{n(out,CL)}}$$

Input referred: divide-by-gain

$$v_{n(in,CL)} = v_{n1} + \frac{1}{A_2} v_{n2}$$



Same input referred noise.

**Feedback does not change
input referred noise density.**

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Equivalent-Input Noise Generators (continued)

- To extract equivalent input voltage noise, use a generator with zero impedance (that is, ideal voltage source).
- To extract equivalent input current noise, use a generator with infinite impedance (that is, ideal current source).
- Only together do the noise current *and* voltage give the full picture for any generator impedance.
- For highest accuracy, sometimes the correlation between current and voltage noise needs to be taken into account.
- Both current noise and voltage noise are usually frequency dependent, for example, due to flicker noise or due to filter effects.
- Often one noise source is dominant, for example, at low frequencies in CMOS opamps usually voltage noise is dominant. Therefore, for simpler calculations it is good enough to use only one kind of noise source.



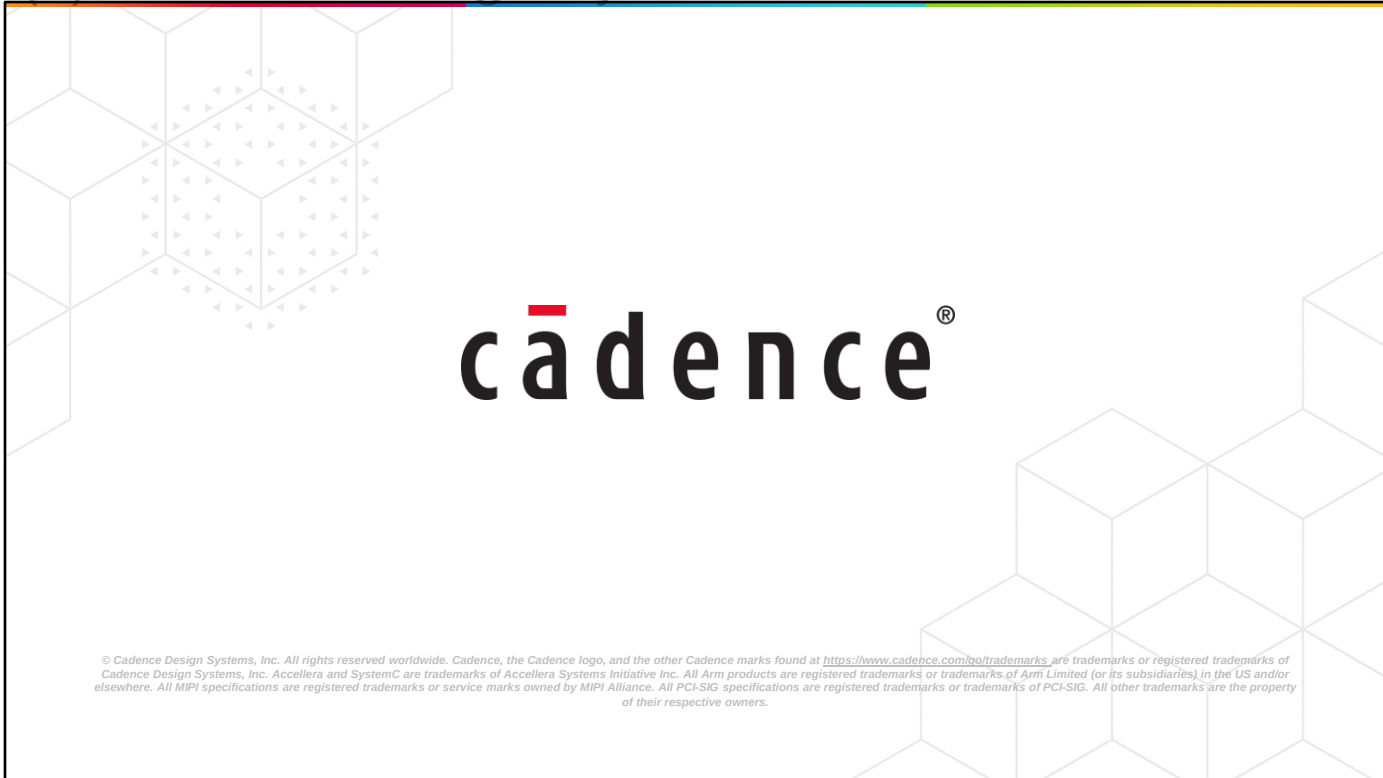
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Summary: Representation of Noise in Circuits

- Noise models are small-signal models in which noisy devices are replaced by ideal devices and a noise source.
- To analyze noise, the noise from all contributors is added together to obtain the noise power.
- In many cases, the noise is referred to the input of the circuit. The minimum level of the input signal that can be processed without affecting its quality is set by the equivalent input referred noise.



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